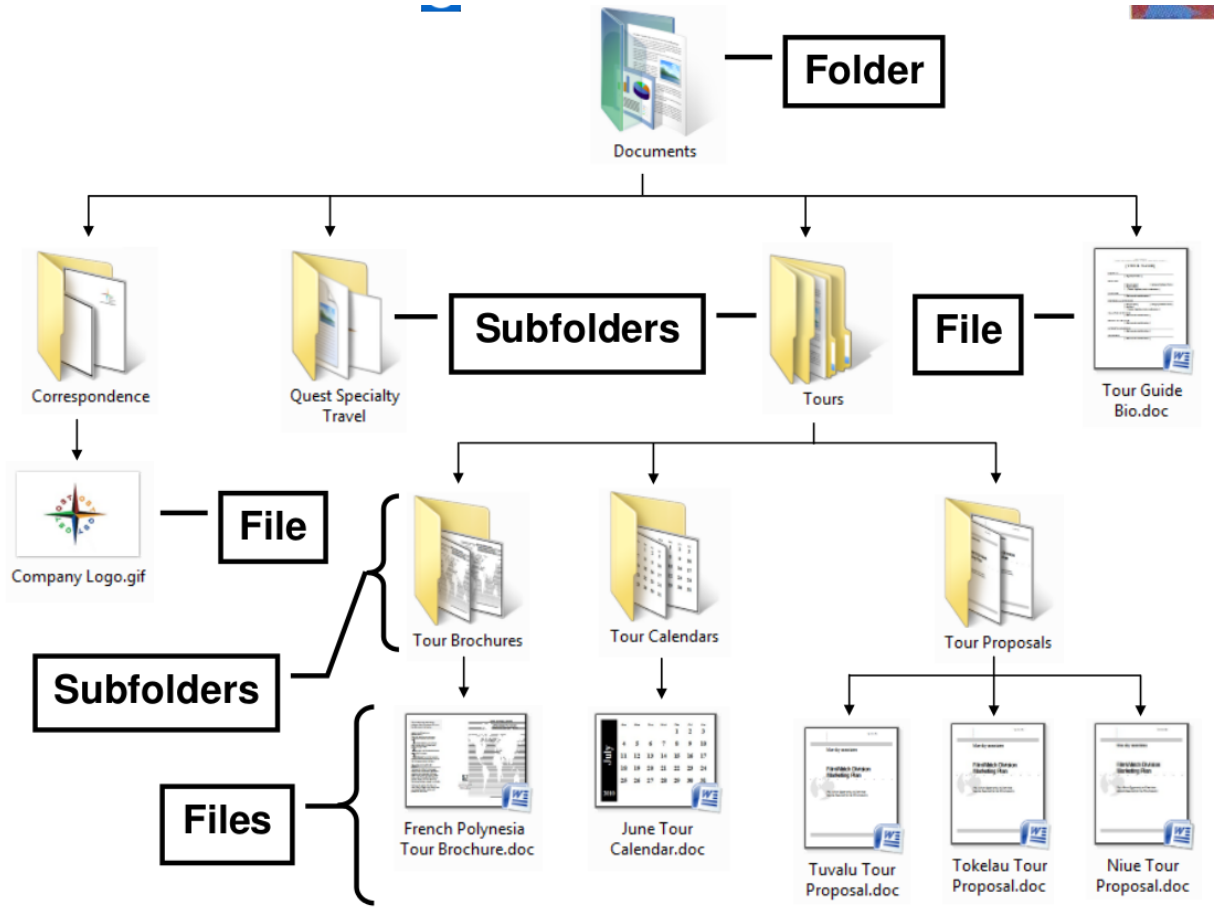
Introduction to UNIX-like systems

Intro. to Computing

`https://mandarm.github.io/computing.html`

File system structure

Files are organised in a hierarchical structure of folders, sub-folders, and files.



Courtesy: `https://www.slideshare.net/okmomwalking/windows-7-unit-b-ppt`

File system structure: terminology

- Folders $\equiv$ *directories*
- Top of the hierarchy: *root directory* (/)
- Default starting location: *home directory* (~)
- Location of a file or directory: specified by *path*
- Current location in terminal or file browser: *current working directory*
- Paths: *absolute* or *relative*
  - absolute path: from root (starts with /)
    Example: `/usr/bin/python3`, `/tmp`, `/home/student`
  - relative path: from current working directory (does not start with /)
    Example: `Desktop/wsl`

Note the difference between (forward) slash (/, used in Unix-like systems) and backslash (\, used in Windows-like systems) !

Navigating the file system

- `cd` : change directory
  Examples:
  `cd` ⟵ go to home directory
  `cd /home/student` (your home directory, same as above)
  `cd /mnt/c/Users/`
- `pwd` : print current working directory

Navigating the file system: special directory names

- ~ : home directory
  Example: `cd ~/Desktop`
- . (dot) : current working directory
  Example: `./program1`
- .. (dot dot) : parent directory (one level up)
  Example: `cd ..`, `cd ../assignment2`, `cd ../..`
- – (dash) : go back to the immediately previous location
  Example:

```
cd /home/student
cd /mnt/c/Users
cd CSSC          use your own username instead of CSSC
cd -             check location using pwd
cd -             check location again using pwd
```

Managing files from the terminal

Listing files

- `ls` : view list of files in current directory
- `ls <path>` : view list of files in specified path
- `ls -l` : view detailed list of files
- `ls -lt` : view detailed list of files sorted by modification time
- `ls -ltr` : view detailed list of files sorted by modification time *in reverse order*

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Example:

```
$ /bin/ls -l
total 68
drwx------ 2 mandar mandar  4096 Jul 19 00:45 assignments
drwx------ 2 mandar mandar  4096 Jul 22  2016 exams
-rw-r--r-- 1 mandar mandar 13521 Jul 19 00:41 index.html
drwx------ 2 mandar mandar  4096 Jul 19 00:45 lectures
```

Managing files from the terminal

- `cp` : copy a file

```
cp program1.c program2.c
cp -i source-file target-file
cp -i source-file target-directory
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```

`-i` ≡ interactive (asks for confirmation)

- `rm` : remove (delete) a file

```
rm program1.c
rm -i file1 file2.c *.bak
```

`rm -r some-directory` (remove directory and everything inside it)

`rm -rf some-directory` (remove directory and everything inside it; dangerous!)

Creating / deleting directories

- `mkdir` : create a directory
  Examples:
  ```
  mkdir clab
  mkdir clab/assignment1
  ```
  Create directories as appropriate.
  OR
  ```
  mkdir -p clab/assignment1 clab/assignment2
  ```
- `rmdir` : remove an (empty) directory
  Example: `rmdir assignment2`, `rmdir clab/programs`

Convenience features

Tab completion: type a part of a command / file name, and hit tab to complete or get a list of suggestions.

Aliases

- **What are they?** → user-defined easy-to-remember names for commands (+ arguments)

  Examples:

```
alias c="clear -x"
alias cp="cp -i"
alias l="ls -l"
alias ls="ls -F"
alias lt="ls -lt"
alias ltr="ls -ltr"
alias mv="mv -i"
alias rm="rm -i"
```

- **Where to define them?** → in `~/.bash_aliases`
- **How does one create / edit files?** → `notepad.exe bash_aliases`

  ("simple and stupid" method that works only on WSL)

Viewers / pagers

- Useful for quickly viewing a file (not editing)
- Use `less` or `bat`

  Example: `less ~/.bash_aliases`
  - space: move forward one page
  - backspace or b: move backward one page
  - q : exit the pager
  - / : search for a string in the file
  - run `man less` for more information

Input/output redirection

Terminology

- "standard input" or *stdin* $\equiv$ input device, i.e., keyboard
- "standard output" or *stdout* $\equiv$ output device, i.e., terminal

Redirection

- Taking input from a **file**: `python3 prog1.py < input.txt`
- Printing stdout to a **file**: `python3 ./prog1 > output.txt`
- Taking stdin / printing stdout from / to a **program**:
  `python3 prog1.py | less`
  vertical bar / pipe character (|)
- Input/output from/to file / program may be combined

```
python prog1.py < input.txt | diff - desired-output.txt > differences.txt
```

Other commands

Find out more about these on your own.

- `cmp`, `diff` (comparing two files)
- `grep` (searching for patterns)
- `awk`, `sed` (programming)
- `wc` (counting characters, words, lines)
- `sort`
- `head`, `tail` (first few / last few lines)
- `ps`, `top`, `kill` (checking what programs are running)
- `find` (finding files or directories)

Useful references / cheat-sheets

http://cli.learncodethehardway.org/bash_cheat_sheet.pdf
https://ubuntudanmark.dk/filer/fwunixref.pdf
http://www.ucs.cam.ac.uk/docs/leaflets/u5
http://mally.stanford.edu/~sr/compuGng/basic-unix.html
http://www.math.utah.edu/lab/unix/unix-commands.html